

**AIMS**African Institute for
Mathematical Sciences
RWANDA

Introduction to Probability and Statistics

Lecture 6: Some Classical Continuous distributions

evans.gouno@univ-ubs.fr

1 Uniform

2 Exponential

3 Gamma

4 Normal

Uniform distribution

Definition

A r.v. X follows a uniform distribution on the domain $[a, b] \subset \mathbb{R}$ iff its p.d.f. is of the form:

$$f(x) = \frac{1}{b-a} \mathbb{1}_{[a,b]}(x)$$

- Notation: $X \sim \mathcal{U}(a, b)$.



Uniform distribution

Definition

A r.v. X follows a uniform distribution on the domain $[a, b] \subset \mathbb{R}$ iff its p.d.f. is of the form:

$$f(x) = \frac{1}{b-a} \mathbb{1}_{[a,b]}(x)$$

- Notation: $X \sim \mathcal{U}(a, b)$.
- Expectation:



Uniform distribution

Definition

A r.v. X follows a uniform distribution on the domain $[a, b] \subset \mathbb{R}$ iff its p.d.f. is of the form:

$$f(x) = \frac{1}{b-a} \mathbb{1}_{[a,b]}(x)$$

- Notation: $X \sim \mathcal{U}(a, b)$.
- Expectation: $E(X) = \frac{1}{b-a} \int_a^b x dx = \frac{a+b}{2}$.
- Variance:

Uniform distribution

Definition

A r.v. X follows a uniform distribution on the domain $[a, b] \subset \mathbb{R}$ iff its p.d.f. is of the form:

$$f(x) = \frac{1}{b-a} \mathbb{1}_{[a,b]}(x)$$

- Notation: $X \sim \mathcal{U}(a, b)$.
- Expectation: $E(X) = \frac{1}{b-a} \int_a^b x dx = \frac{a+b}{2}$.
- Variance: $E(X^2) = \frac{1}{b-a} \int_a^b x^2 dx = \frac{1}{b-a} \frac{b^3 - a^3}{3} = \frac{a^2 + ab + b^2}{3}$ and

$$\text{Var}(X) = E(X^2) - (EX)^2 = \frac{(b-a)^2}{12}.$$

Uniform distribution

Definition

A r.v. X follows a uniform distribution on the domain $[a, b] \subset \mathbb{R}$ iff its p.d.f. is of the form:

$$f(x) = \frac{1}{b-a} \mathbb{1}_{[a,b]}(x)$$

- Notation: $X \sim \mathcal{U}(a, b)$.
- Expectation: $E(X) = \frac{1}{b-a} \int_a^b x dx = \frac{a+b}{2}$.
- Variance: $E(X^2) = \frac{1}{b-a} \int_a^b x^2 dx = \frac{1}{b-a} \frac{b^3 - a^3}{3} = \frac{a^2 + ab + b^2}{3}$ and

$$\text{Var}(X) = E(X^2) - (EX)^2 = \frac{(b-a)^2}{12}.$$

Exponential distribution

Definition

A r.v. X has an exponential distribution with parameter $\theta > 0$ iff its p.d.f. has the form:

$$f(x | \theta) = \frac{1}{\theta} \exp \left\{ -\frac{x}{\theta} \right\}, \quad x \in \mathbb{R}^+.$$

- Notation: $X \sim \mathcal{E}(\theta)$
- Cumulative distribution function: $F(x) = 1 - e^{-x/\theta}$.
- Moment generating function:

Exponential distribution

Definition

A r.v. X has an exponential distribution with parameter $\theta > 0$ iff its p.d.f. has the form:

$$f(x | \theta) = \frac{1}{\theta} \exp \left\{ -\frac{x}{\theta} \right\}, \quad x \in \mathbb{R}^+.$$

- Notation: $X \sim \mathcal{E}(\theta)$
- Cumulative distribution function: $F(x) = 1 - e^{-x/\theta}$.
- Moment generating function: $M_X(t) = \frac{1}{1 - \theta t}$.

Exponential distribution

Definition

A r.v. X has an exponential distribution with parameter $\theta > 0$ iff its p.d.f. has the form:

$$f(x | \theta) = \frac{1}{\theta} \exp \left\{ -\frac{x}{\theta} \right\}, \quad x \in \mathbb{R}^+.$$

- Notation: $X \sim \mathcal{E}(\theta)$
- Cumulative distribution function: $F(x) = 1 - e^{-x/\theta}$.
- Moment generating function: $M_X(t) = \frac{1}{1 - \theta t}$.
- Expectation:

Exponential distribution

Definition

A r.v. X has an exponential distribution with parameter $\theta > 0$ iff its p.d.f. has the form:

$$f(x | \theta) = \frac{1}{\theta} \exp \left\{ -\frac{x}{\theta} \right\}, \quad x \in \mathbb{R}^+.$$

- Notation: $X \sim \mathcal{E}(\theta)$
- Cumulative distribution function: $F(x) = 1 - e^{-x/\theta}$.
- Moment generating function: $M_X(t) = \frac{1}{1 - \theta t}$.
- Expectation: $E(X) = \theta$.

Exponential distribution

Definition

A r.v. X has an exponential distribution with parameter $\theta > 0$ iff its p.d.f. has the form:

$$f(x | \theta) = \frac{1}{\theta} \exp \left\{ -\frac{x}{\theta} \right\}, \quad x \in \mathbb{R}^+.$$

- Notation: $X \sim \mathcal{E}(\theta)$
- Cumulative distribution function: $F(x) = 1 - e^{-x/\theta}$.
- Moment generating function: $M_X(t) = \frac{1}{1 - \theta t}$.
- Expectation: $E(X) = \theta$.
- Variance:

Exponential distribution

Definition

A r.v. X has an exponential distribution with parameter $\theta > 0$ iff its p.d.f. has the form:

$$f(x | \theta) = \frac{1}{\theta} \exp \left\{ -\frac{x}{\theta} \right\}, \quad x \in \mathbb{R}^+.$$

- Notation: $X \sim \mathcal{E}(\theta)$
- Cumulative distribution function: $F(x) = 1 - e^{-x/\theta}$.
- Moment generating function: $M_X(t) = \frac{1}{1 - \theta t}$.
- Expectation: $E(X) = \theta$.
- Variance: $\text{Var}(X) = \theta^2$.

Gamma distribution

Definition

A r.v. X has a gamma distribution with parameter $\alpha, \beta > 0$ iff its p.d.f. has the form:

$$f(x \mid \alpha, \beta) = \frac{\beta^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\beta x}, \quad x \in \mathbb{R}^+$$

where $\Gamma(\cdot)$ is the gamma function defined by: $\Gamma(\alpha) = \int_0^{+\infty} u^{\alpha-1} e^{-u} du$.

- Notation: $X \sim \mathcal{G}(\alpha, \beta)$
- Expectation: $E(X) = \alpha/\beta$.
- Variance: $\text{Var}(X) = \alpha/\beta^2$.
- Moment generating function: $M_X(t) = \left(\frac{\beta}{\beta - t} \right)^\alpha$.

Gamma distribution

Exercise

Let $X_1, X_2, \dots, X_n$ be n independent exponential r.v with parameter θ .

Find the distribution of $S_n = \sum_{i=1}^n X_i$ using the moment generating function.

Normal (Gaussian) distribution



Normal (Gaussian) distribution

Definition

A r.v. X has a normal with parameters (μ, σ^2) $\mu \in \mathbb{R}$, $\sigma^2 > 0$ iff its p.d.f. is:

$$f(x \mid \mu, \sigma^2) = \frac{1}{\sqrt{2\pi}\sigma} \exp \left\{ -\frac{(x - \mu)^2}{2\sigma^2} \right\}, \quad x \in \mathbb{R}.$$

- Notation: $X \sim \mathcal{N}(\mu, \sigma^2)$.

Normal (Gaussian) distribution

Definition

A r.v. X has a normal with parameters (μ, σ^2) $\mu \in \mathbb{R}$, $\sigma^2 > 0$ iff its p.d.f. is:

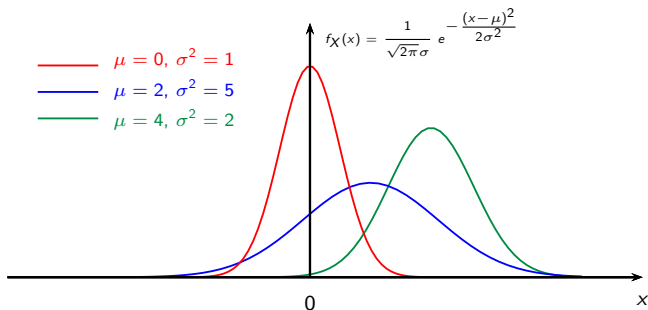
$$f(x \mid \mu, \sigma^2) = \frac{1}{\sqrt{2\pi}\sigma} \exp \left\{ -\frac{(x - \mu)^2}{2\sigma^2} \right\}, \quad x \in \mathbb{R}.$$

- Notation: $X \sim \mathcal{N}(\mu, \sigma^2)$.
- Cumulative distribution function:

$$F(x) = \int_{-\infty}^x \frac{1}{\sqrt{2\pi}\sigma} \exp \left\{ -\frac{(t - \mu)^2}{2\sigma^2} \right\} dt.$$

Normal (Gaussian) distribution

p.d.f. of a normal r.v for different parameters values



Normal (Gaussian) distribution



Normal distribution

- When $\mu = 0$ and $\sigma^2 = 1$, X is said to follow the **standard normal distribution**.
- p.d.f. of the standard normal distribution:

$$f(x \mid 0, 1) = \phi(x) = \frac{1}{\sqrt{2\pi}} \exp \left\{ -\frac{x^2}{2} \right\}, \quad x \in \mathbb{R}.$$

Normal distribution

- When $\mu = 0$ and $\sigma^2 = 1$, X is said to follow the **standard normal distribution**.
- p.d.f. of the standard normal distribution:

$$f(x \mid 0, 1) = \phi(x) = \frac{1}{\sqrt{2\pi}} \exp \left\{ -\frac{x^2}{2} \right\}, \quad x \in \mathbb{R}.$$

- c.d.f. of the standard normal distribution:

$$\Phi(x) = \int_{-\infty}^x \frac{1}{\sqrt{2\pi}} \exp \left\{ -\frac{t^2}{2} \right\} dt.$$

Normal distribution

- When $\mu = 0$ and $\sigma^2 = 1$, X is said to follow the **standard normal distribution**.
- p.d.f. of the standard normal distribution:

$$f(x \mid 0, 1) = \phi(x) = \frac{1}{\sqrt{2\pi}} \exp \left\{ -\frac{x^2}{2} \right\}, \quad x \in \mathbb{R}.$$

- c.d.f. of the standard normal distribution:

$$\Phi(x) = \int_{-\infty}^x \frac{1}{\sqrt{2\pi}} \exp \left\{ -\frac{t^2}{2} \right\} dt.$$

Normal distribution

Checking that φ is a pdf that is $\int_{\mathbb{R}} \frac{1}{\sqrt{2\pi}} e^{-\frac{z^2}{2}} dz = 1$.

Let $I = \int_{\mathbb{R}} e^{-\frac{z^2}{2}} dz$ and compute I^2 .

$$I^2 = \left(\int_{\mathbb{R}} e^{-\frac{u^2}{2}} du \right) \left(\int_{\mathbb{R}} e^{-\frac{v^2}{2}} dv \right) = \int_{\mathbb{R}} \int_{\mathbb{R}} e^{-\frac{u^2+v^2}{2}} dudv.$$

Making the change of variables $u = r \cos \theta$ and $v = r \sin \theta$, we have $du dv = r dr d\theta$. The integration domain becomes $[0, +\infty[\times [0, 2\pi]$.

$$\begin{aligned} I^2 &= \int_0^{+\infty} \int_0^{2\pi} e^{-r^2/2} r dr d\theta = \left(\int_0^{2\pi} d\theta \right) \times \left(\int_0^{+\infty} r e^{-r^2/2} dr \right) \\ &= \left[-e^{-r^2/2} \right]_0^{+\infty} = 2\pi. \end{aligned}$$

Therefore $I = \sqrt{2\pi}$.

Normal distribution

Proposition

$$(X \sim \mathcal{N}(\mu, \sigma^2)) \iff \left(Z = \frac{X - \mu}{\sigma} \sim \mathcal{N}(0, 1) \right).$$

Proof: Express $P(X \leq x)$. Then $P(Z \leq z)$.

Normal distribution

Proof: Suppose $X \sim \mathcal{N}(\mu, \sigma^2)$. What is the distribution of $Z = (X - \mu)/\sigma$?

$$P(Z \leq z) = P\left(\frac{X - \mu}{\sigma} \leq z\right) = P(X \leq \mu + \sigma z) = \int_{-\infty}^{\mu + \sigma z} \frac{1}{\sqrt{2\pi}\sigma} \exp\left\{-\frac{(t - \mu)^2}{2\sigma^2}\right\} dt.$$

Let's set $u = (t - \mu)/\sigma$. Then $t = \sigma u + \mu$, $dt = \sigma du$ and when t varies between $-\infty$ and $\mu + \sigma z$, u varies between $-\infty$ and z .

Thus

$$P(Z \leq z) = \int_{-\infty}^z \frac{1}{\sqrt{2\pi}} \exp\left\{-\frac{u^2}{2}\right\} du = \Phi(z).$$

Conversely, consider $Z \sim \mathcal{N}(0, 1)$. What is the distribution of X such that $Z = (X - \mu)/\sigma$ i.e. the distribution of $X = \sigma Z + \mu$.

$$P(X \leq x) = P(\sigma Z + \mu \leq x) = P\left(Z \leq \frac{x - \mu}{\sigma}\right) = \int_{-\infty}^{(x - \mu)/\sigma} \frac{1}{\sqrt{2\pi}} \exp\left\{-\frac{u^2}{2}\right\} du.$$

Let's set $u = (t - \mu)/\sigma$. The $du = dt/\sigma$ and when u is in $] -\infty, (x - \mu)/\sigma]$, t is $] -\infty, x]$.

Thus

$$P(X \leq x) = \int_{-\infty}^x \frac{1}{\sqrt{2\pi}} \exp\left\{-\frac{(t - \mu)^2}{2\sigma^2}\right\} \frac{dt}{\sigma}.$$

We recognize the normal distribution with parameters (μ, σ^2) .

Normal distribution

Exercise

Compute the moment generating function (m.g.f.) of a standard normal r.v.

Find the m.g.f of a normal r.v. with parameters (μ, σ^2) using the previous proposition.

Normal distribution

Exercise

Compute the moment generating function (m.g.f.) of a standard normal r.v.

Find the m.g.f of a normal r.v. with parameters (μ, σ^2) using the previous proposition.

$$\left(X \sim \mathcal{N}(\mu, \sigma^2) \right) \iff \left(M_X(t) = \exp \left\{ \mu t + \frac{\sigma^2}{2} t^2 \right\} \right).$$

Normal distribution

- **Expectation**

Let $Z \sim \mathcal{N}(0, 1)$, then $E(Z) = 0$.

Let $X = \mu + \sigma Z$, $X \sim \mathcal{N}(\mu, \sigma^2)$ and $E(X) = \mu$.

- **Variance**

$$\text{Var}(Z) = E(Z^2) - (EZ)^2.$$

It can be shown that: $E(Z^2) = 1$. Thus: $\text{Var}(Z) = 1$.

$$\text{Var}(X) = \text{Var}(\mu + \sigma Z) = \sigma^2 \text{Var}(Z) = \sigma^2.$$

- **Conclusion**

If $X \sim \mathcal{N}(\mu, \sigma^2)$, then $E(X) = \mu$ and $\text{Var}(X) = \sigma^2$.

Normal distribution

Proposition

$$\forall z, \Phi(z) + \Phi(-z) = 1.$$

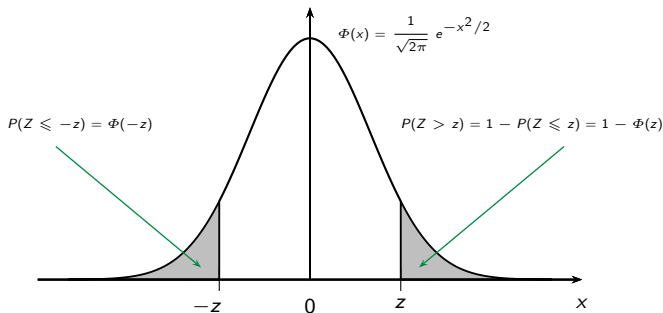
Proof: compute $P(Z \leq -z)$.

Normal distribution

Proposition

$$\forall z, \Phi(z) + \Phi(-z) = 1.$$

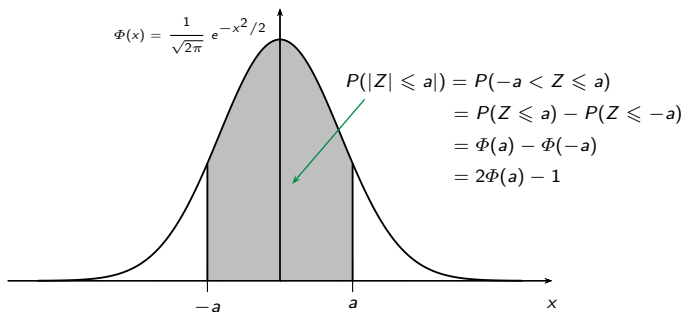
Proof: compute $P(Z \leq -z)$.



Normal distribution

Proposition

Soit $Z \sim \mathcal{N}(0, 1)$, $P(|Z| \leq a) = 2\Phi(a) - 1$.



Normal distribution

Remark: $P(|Z| > a) = 1 - P(|Z| \leq a) = 1 - 2\Phi(a) + 1 = 2(1 - \Phi(a))$.

Let $P(|Z| \leq a) = \gamma$, then $\Phi(a) = \frac{1 + \gamma}{2}$.

The quantile a associated with the probability γ is $a = \Phi^{-1}\left(\frac{1 + \gamma}{2}\right)$.

Normal distribution

Let consider $X \sim \mathcal{N}(\mu, \sigma^2)$.

We have:

$$P\left(\left|\frac{X - \mu}{\sigma}\right| \leq a\right) = \gamma \iff P(\mu - a\sigma < X \leq \mu + a\sigma) = \gamma.$$

Therefore the probability that the r.v. lies in an interval of length $2a\sigma$ centered on μ is equal to γ .

- For $a = 1$, $\Phi(1) = 0.8413$ and $\gamma = 2 \times 0.8413 - 1 = 0.6826$.
- For $a = 2$, $\Phi(2) = 0.9772$ and $\gamma = 2 \times 0.9772 - 1 = 0.9544$.
- For $a = 3$, $\Phi(3) = 0.9986$ and $\gamma = 2 \times 0.9986 - 1 = 0.9973$.